

Esau Salvador Martinez Lopez

AI Engineer

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Profile Summary

AI Engineer with 4+ years building production AI systems for government and legal tech. Currently leading R&D at the State Prosecutor's Office of Guanajuato, Mexico, where forensic tools I developed are being evaluated by multiple jurisdictions. Previously built AI-powered workflows serving 1,000+ customers at a US legal tech company.

Professional Experience

AI Engineer

May 2026 – Present – On-site

State Prosecutor's Office of Guanajuato, Mexico

- Lead R&D initiatives developing AI-powered forensic tools. Manage an on-premise cluster of 8 NVIDIA H100 GPUs with vLLM for model deployment, KV-cache optimization, and performance tuning.
- Now heading the technical direction of the forensic AI portfolio, overseeing ongoing projects and adoption efforts.
- Co-developed **“Rostros que no se olvidan”**, a forensic aging system using computer vision that achieves 94% accuracy with 248+ progressions. In evaluation by 4 additional state prosecutor offices.
- Co-developed a **tattoo-based person search system** using image embeddings for forensic identification in SEMEFO (forensic services).
- Developed a **phone records analysis platform** with interactive statistics and maps for investigators.

Backend Software Engineer

Dec 2021 – Mar 2026 – Remote

Justia — Legal Technology (US)

- Designed backend services in TypeScript (Node.js) and AWS Lambda for large-scale legal automation, serving 1,000+ customers.
- Built a **semantic search system** using vector embeddings and AI vector databases, enabling relevance-based document retrieval.
- Developed an **AI Writer pipeline** using LangChain and LangGraph, orchestrating multiple LLMs (OpenAI, Gemini) for structured content creation.
- Built an **AI-driven legal chatbot agent** using LangChain and LangGraph for structured question creation and legal domain classification.
- Designed event-driven pipelines using Redis-backed queues and AWS Step Functions for multi-stage workflow orchestration.

Technical Skills

AI & LLMs: OpenAI, Gemini, LangChain, LangGraph, prompt engineering, multi-LLM orchestration, AI agents, machine learning, computer vision

ML Infrastructure: NVIDIA H100 GPU clusters, model deployment, PyTorch, HuggingFace, MLOps

Computer Vision: facial aging models, image embeddings, semantic similarity search, tattoo recognition, OpenCV

Cloud & Infrastructure: AWS (Lambda, Step Functions, EC2, S3), Docker, Kubernetes, CI/CD, microservices

Backend & Frameworks: Node.js, Express.js, AdonisJS, REST APIs

Data Stores: MySQL, Redis (queues, caching), Amazon Redshift (columnar analytics)

Programming Languages: TypeScript, JavaScript (ES6+), Python

Education

Universidad Tecnológica de León

2018 – 2023

B.S. in Software Engineering

Université Grenoble Alpes

2020 – 2021

International Exchange Program — Software Engineering

Languages

Spanish: Native **English:** C1 **French:** B2